Case 07: Ignition Probability and SFT Convergence

1 Verification purpose

This case verifies statistical convergence of the stochastic smoldering-to-flaming (SFT) model and the resulting ignition-probability response. Pooled ignition-delay samples must converge to the configured SFT scale, while ensemble-mean firebrand-driven spread must respond consistently to ignition probability.

The test exposes biased sampling, insufficient ensemble size, incorrect delay extraction, and grid-dependent probability application. Success is determined from confidence-aware delay statistics and declared ensemble metrics, never from a single realization.

2 Mathematical formulation and numerical implementation

For accumulated firebrands in a cell, ELMFIRE's SIMPLE model uses

$$F(t) = 1 - (1 - P)^{t/\tau}, \qquad P_{\Delta t} = 1 - (1 - P)^{\Delta t/\tau}, \tag{1}$$

where $P = P_{\text{ign,small}}$, $\tau = \tau_{\text{ign}}$, and $P_{\Delta t}$ is the Bernoulli probability evaluated during one solver step. Consequently $F(\tau) = P$: at $P = 0.9$, the theoretical 90th percentile is $q_{0.9} = \tau = 10$ s. A successful local transition is followed by additional development delay=0, so the final time of arrival (TOA) records the full ignition time without an additional development lag.

For the dedicated delay members, transient ember-flux output is written every wind Courant–Friedrichs–Lewy (CFL) step. If a cell's first nonzero transient flux occurs at dump k, its first deposition lies in $(t_{k-1}, t_k]$. Postprocessing uses the interval midpoint $t_{d,i} = (t_{k-1} + t_k)/2$ and forms the actual-output sample

$$d_i = t_{\text{ign},i} - t_{d,i}. \tag{2}$$

Only centerline cells marked in the final ember-ignition raster, outside the two-cell halo and in the 155–2800 m complete-support interval, are admitted. The dump spacing Δt is retained as the temporal-censoring resolution.

For the response sweep,

$$CFL_w = \frac{u_w \Delta t}{\Delta x} \leq 0.5, \qquad r_m = \frac{R_m}{u_w}, \tag{3}$$

and each member rate of spread (ROS) R_m is the least-squares slope of centerline distance against final TOA over 100–2800 m. Ten explicit seeds are paired across $\Delta x = 5, 10, 20, 30$ m and $P = 0, 0.25, 0.5, 0.75, 1$.

3 Derivation of expected behavior

Equation 1 gives the exact event-level expectation $q_{0.9} = 10$ s. A finite sample quantile fluctuates, but its bootstrap confidence interval should contract as independently seeded members add cell samples. The final pooled estimate should therefore approach 10 s rather than an ROS-derived surrogate. Timestep-resolved deposition limits the deterministic timing uncertainty to the deposition interval.

At $P = 0$, the ignition hazard is zero, and no spotting-driven front should propagate. Increasing P increases $F(t)$ for every $t > 0$ and therefore cannot reduce population-mean ROS. At $P = 1$, ignition

is immediate in the continuous model and $\overline{r}$ should approach one. Because physical extent is fixed and the generated CFL_w values remain conservatively bounded near 0.5, the four resolution curves should nearly coincide.

For the delay member, $P = 0.9$ and $\tau = 10$ s, hence

$$F(5) = 1 - 0.1^{0.5} = 0.683772, \qquad F(10) = 1 - 0.1 = 0.9, \qquad F(20) = 1 - 0.1^2 = 0.99.$$

Solving $F(q_{0.9}) = 0.9$ gives $0.1^{q_{0.9}/10} = 0.1$ and therefore $q_{0.9} = 10$ s exactly. With $U = 15(0.44704) = 6.7056\ \mathrm{m\,s^{-1}}$, preprocessing first computes the $CFL_w = 0.5$ upper bounds, then reduces each step to divide 240 s exactly. The actual generated steps and corresponding per-step ignition probabilities are

Δx (m)	5	10	20	30
Δt (s)	0.372671	0.745342	1.490683	2.222222
$1 - 0.1^{\Delta t/10}$	0.082232	0.157702	0.290534	0.400516

For each resolution the conservative CFL-derived timestep defines the probability transformation and cross-resolution control. Preprocessing selects $N = \lceil 240/\Delta t_{CFL} \rceil$, writes the conservative divisor $\Delta t = 240/N$, and keeps $t_{stop} = 240$ s exactly; the simulation specification records nominal and actual CFL values plus the integer step count. This shows why the step probability changes while the continuous delay cumulative distribution function (CDF) does not.

The response design contains 4 resolutions $\times 5$ probabilities $\times 10$ paired seeds $= 200$ required runs, plus 10 delay members. The exact limiting references are $R/U = 0$ at $P = 0$, monotone nondecrease with P, and $R/U \to 1$ at $P = 1$. Numerically these become a maximum cross-resolution range of 0.10, a $P = 1$ floor of 0.90, and adjacent decreases no larger than their pooled 95-percent confidence half-width. The delay substitution is $|\widehat{q}_{0.9} - 10| \leq 1$ s, with at least 1000 samples and bootstrap half-width at most 1 s.

4 Verification metrics and acceptance criteria

For every response group with $N = 10$, the postprocessor calculates the mean, sample standard deviation, standard error, and two-sided Student-t 95% confidence interval. Adjacent probability means may decrease only within their pooled confidence half-widths. At each P, the spread of the four resolution means must not exceed 0.10, and every $P = 1$ mean must be at least 0.90.

For the delay ensemble, samples are pooled after each independent member. At the final member, all ten members must be usable, at least 1000 delays must be available, $|\widehat{q}_{0.9} - 10\ \mathrm{s}| \leq 1$ s, and the 2000-replicate bootstrap 95% interval half-width must not exceed 1 s. A missing final TOA, ignition map, dump-time table, transient sequence, stale raster shape, or empty sample set makes that member incomplete. Overall PASS is the conjunction of all response and delay components; missing required output yields insufficient simulation results, never a pass.

4.1 Justification of metric quantification

A two-sided 95% Student-t interval is used because each response mean has only ten independent ensemble members and its variance is estimated from those members. Allowing an adjacent decrease only within the pooled confidence half-width distinguishes sampling noise from evidence against monotonicity without assuming equal variances. The 0.10 cross-resolution range and the 0.90 unit-probability floor each represent ten percentage points on the naturally normalized $[0, 1]$ response scale: they admit finite-ensemble and grid-crossing variability but reject a practically material resolution effect or failure to approach immediate ignition.

The target $q_{0.9} = 10\,\mathrm{s}$ follows directly from the prescribed SFT distribution [1]. Requiring 1000 pooled delays gives substantial support in the upper decile rather than allowing a few extreme samples to set the quantile. The 1 s accuracy and bootstrap-half-width limits are each 10% of the model timescale, so both bias and uncertainty must be small on the scale being verified. Two thousand bootstrap replicates provide stable percentile endpoints for this engineering decision without treating the bootstrap interval as an exact physical confidence bound. Full member completeness is required because silently dropping failed stochastic runs can bias both response means and delay quantiles.

4.2 Predeclared acceptance criteria

Table 1 summarizes the metrics fixed before inspection of the calculated results. Numerical values and component statuses are reported separately in Section 7.

Table 1: Predeclared verification metrics and acceptance criteria.

Metric	Output, region, and aggregation	Acceptance criterion
Response-ensemble completeness	Ten usable members for every resolution–probability group.	200 of 200 members
Probability monotonicity	Adjacent ensemble-mean normalized ROS differences, allowing decrease only within pooled 95% confidence half-widths.	Every adjacent check passes
Resolution spread	Range of the four resolution means at each ignition probability.	≤ 0.10
Unit-probability response	Minimum ensemble-mean normalized ROS at $P_{ign} = 1$.	≥ 0.90
SFT sample completeness	Ten usable delay members and pooled interval-censored ignition-delay samples.	10 members and at least 1000 samples
SFT 90th percentile	Absolute difference between pooled $\widehat{q}_{0.9}$ and the configured 10 s scale.	$\leq 1\,\mathrm{s}$
SFT uncertainty	Half-width of the 2000-replicate bootstrap 95% interval for $\widehat{q}_{0.9}$.	$\leq 1\,\mathrm{s}$
Overall decision	Conjunction of every response and SFT component; incomplete output is not evaluable.	PASS only if all pass

5 Simulation configuration and predicted outcome

Figure 1 records the prepared spatial inputs for the first ensemble member with a 10 m grid and ignition probability 0.5. These panels show the prepared spatial fields, remove the two-cell numerical buffer, and retain map coordinates in metres. The left panel shows the most informative fuel or structure layout available for the case; the right panel shows the cells selected by the non-positive initial level-set field, making a localized ignition or firebrand-emission source explicit.

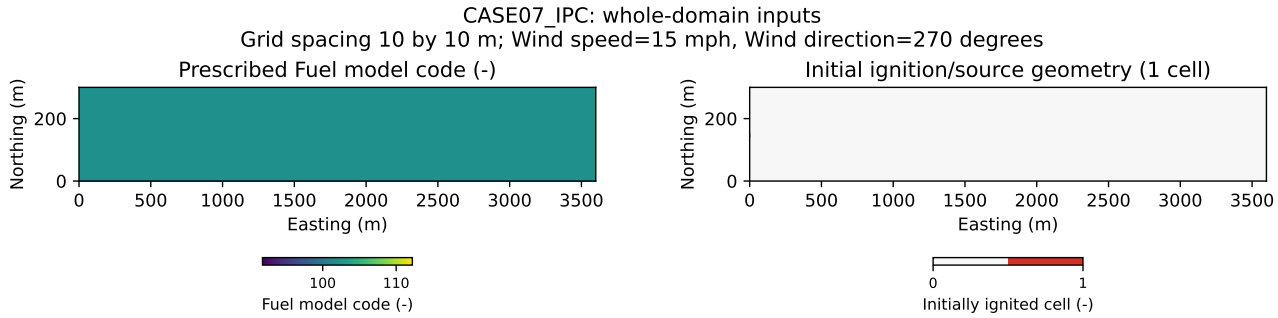


Figure 1: Whole-domain prepared input configuration for the representative simulation condition. The physical domain is shown without the numerical halo; coordinates, configured wind values, fuel or structure layout, and initial ignition/source geometry are identified in the panels.

Parameter	Configured value	Unit	Verification role
Physical domain	3600×300	m	Fixed physical extent
Buffer	2 cells per side	cells	Excluded numerical halo
Response grids	5, 10, 20, 30	m	Resolution comparison
Delay grid	10	m	delay-ensemble spatial baseline
Wind	15 (6.7056)	mph (m/s)	Uniform transport
Wind CFL	≤ 0.5	–	stable near-fixed $\Delta t/\Delta x$
Stop time	240	s	exact timestep multiple
Response probabilities	0, 0.25, 0.50, 0.75, 1	–	probability-response design
Delay model	$P = 0.9$, $\tau = 10$	–, s	delay-distribution design
Ensembles	20×10 response; 10 delay	runs	Sampling statistics
Generation	PER-AREA, 10	pcs m^{-2} s^{-1}	No-surface source
Accumulation/ignition	EULERIAN/SIMPLE	–	Implementation under test
Surface/development delay	disabled/0	–, s	Isolate SFT waiting time
Delay output cadence	one wind-CFL step	s	Bracket first deposition

Table 2: Configuration of the merged ensemble-convergence case verification.

The predicted response plot contains four nondecreasing curves beginning at zero and approaching one. The delay plot should approach the horizontal 10 s line, with its sampling interval narrowing as ensemble members are pooled.

6 Actual simulation results

Figure 2 shows the representative ember-ignition field from the first ensemble member with a 10 m grid and ignition probability 0.5. The displayed field is taken from the simulated results, masks missing values, and excludes the two-cell numerical buffer. This representative ensemble member supplies spatial evidence alongside the ensemble probability and delay statistics; it is not used as a replacement for those statistical metrics.

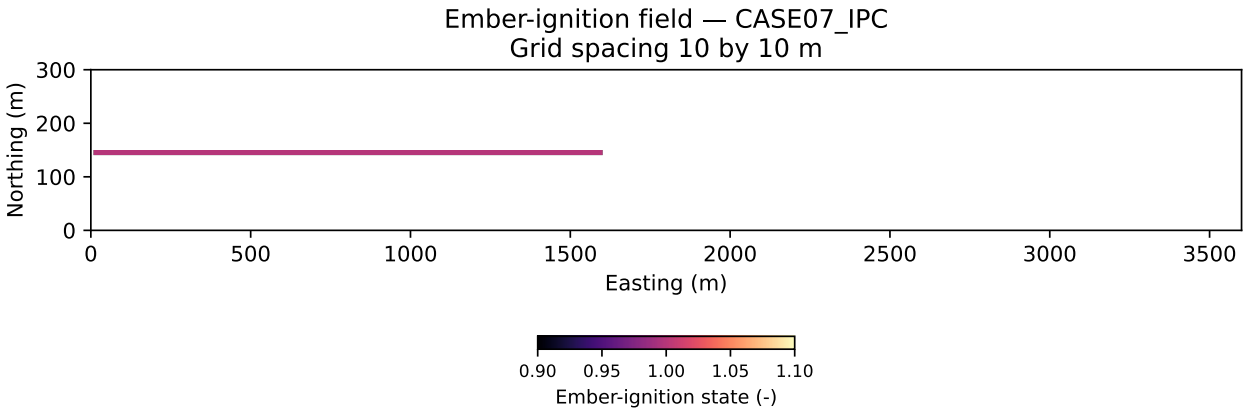


Figure 2: Whole-domain ember-ignition field from the representative completed ELMFIRE run. This domain-scale result supplements, rather than replaces, the higher-level verification profiles, convergence plots, ensemble statistics, and calculated acceptance metrics below.

The distribution-level ignition-delay check is presented first because it compares the sampled stochastic variable directly with its prescribed model quantile. Figure 3 uses only delays reconstructed from actual arrival-time, ember-ignition, dump-time, and transient-flux outputs.

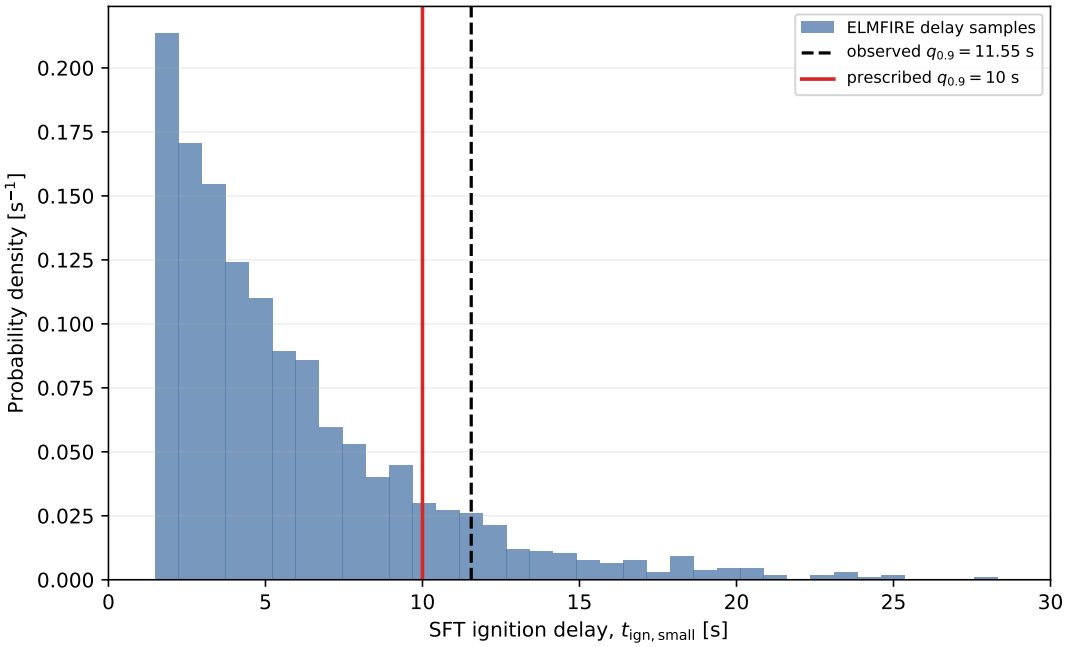


Figure 3: Probability density of sampled SFT ignition delays. The solid red line is the prescribed 90th percentile and the dashed black line is the observed pooled 90th percentile.

Figure 4 presents the second reference observable: the ensemble-mean normalized leading-edge ROS response to ignition probability and spatial resolution.

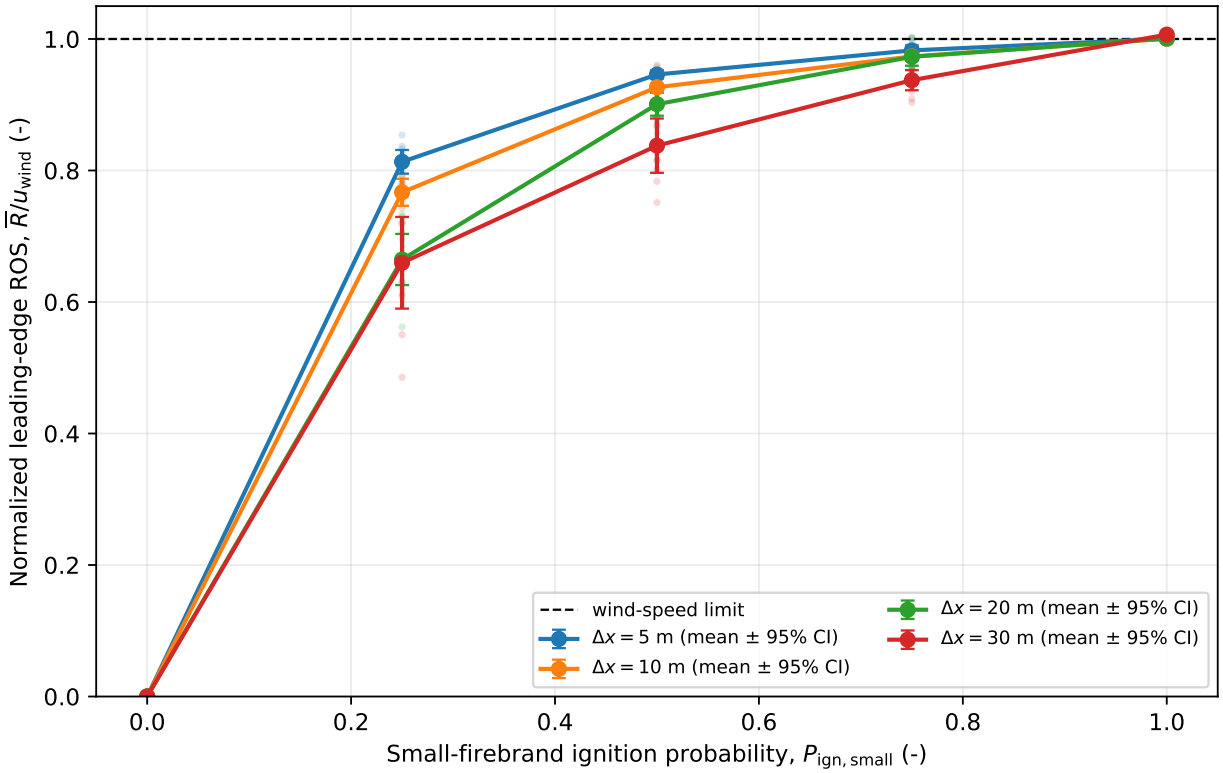


Figure 4: Ensemble-mean normalized leading-edge ROS versus $P_{\text{ign,small}}$. Error bars are 95% confidence intervals and faint points are independently seeded ELMFIRE members.

Figure 5 is an additional sample-size diagnostic. It shows how the pooled delay quantile and its bootstrap uncertainty change as independent members are added.

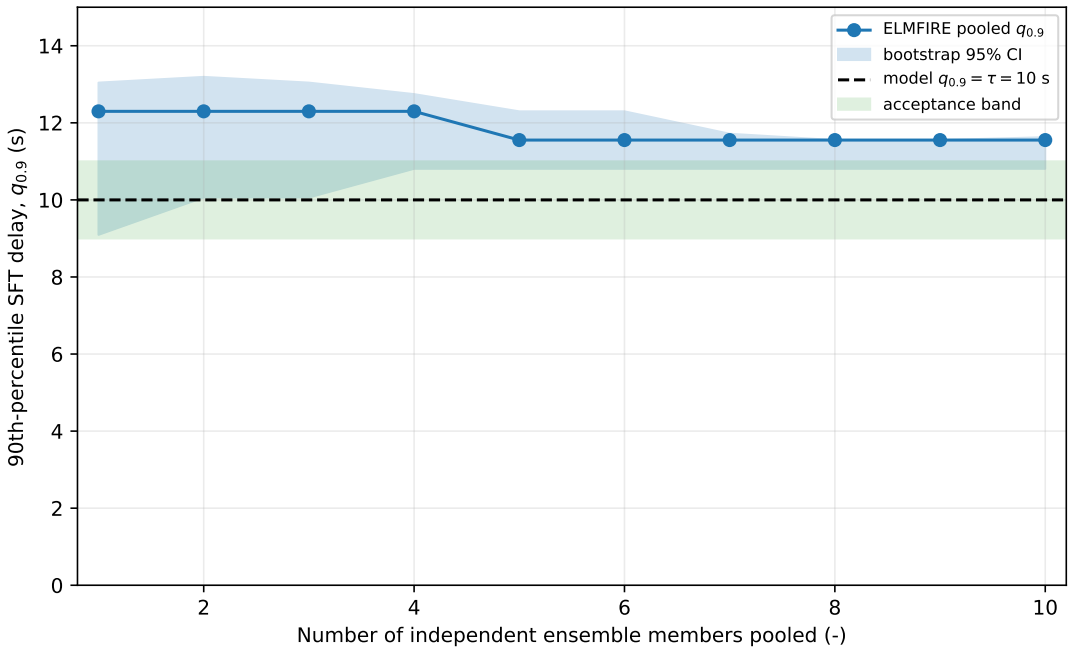


Figure 5: Cumulative convergence of the pooled ELMFIRE 90th-percentile SFT delay. The dashed line is the model expectation, blue shading is the bootstrap 95% interval, and the green band is the ± 1 s acceptance interval.

7 Calculated metrics and verification decision

Metric	Requirement	Calculated
Response members	200 usable	200/200
Probability response	monotone within 95% uncertainty	True
Maximum resolution-mean spread	≤ 0.10	0.1536
Minimum $P = 1$ mean	≥ 0.90	1.0004
SFT members	10 usable	10/10
SFT delay samples	≥ 1000	1440
SFT 90th percentile	10 ± 1 s	11.5528 s
SFT quantile absolute error	≤ 1 s	1.5528 s
SFT bootstrap half-width	≤ 1 s	0.4099 s
SFT component	all delay checks pass	False
Overall status	all components pass	**FAIL**

Table 3: Calculated verification metrics for the completed simulations.

The overall decision is **FAIL**.

8 Technical interpretation

All 200/200 response members and 10/10 dedicated delay members are usable. The maximum cross-resolution response spread is 0.1536, exceeding the 0.10 limit, although the minimum unit-probability mean passes. The 1440 admitted SFT samples give a 90th-percentile delay of 11.5528 s; its 1.5528-s

absolute error exceeds the 1-s limit, while the bootstrap half-width passes. Thus the result is a genuine **FAIL**, not an incomplete evaluation. The generated probability, seed, timestep, and ensemble controls are internally consistent. The source updates probability as $1-(1-P)^{\Delta t/\tau}$ and advances a newly deposited cell through its staged ignition state on later invocations; the observed extra-step delay and grid dependence therefore identify current source timing/discretization behavior rather than a case-configuration defect.

9 Scope and limitations

The case verifies one SIMPLE SFT time scale, one delay probability, one uniform wind and fuel, Eulerian accumulation, zero full-development delay, and the specified grid/probability sweep. Midpoint reconstruction of first deposition is interval-censored at one wind-CFL step; that resolution is recorded but is not sampling uncertainty. Cell samples within a run share a propagating fire history, so independent run count and pooled cell count are both reported. The test does not validate the empirical choice of P or τ, physical ignition thresholds, surface-fire competition, PHYSICAL ignition, Lagrangian transport, or firebrand consumption.

References

[1] Y. Qin, *A Physics-Based Eulerian Framework for Modeling Firebrand Showering in Regional-Scale Wildland and Wildland–Urban Interface Fire Simulations*, Ph.D. dissertation, University of Maryland, College Park, 2025.